S-Parameter Sensitivities for Electromagnetic Optimization Based on Volume Field Solutions

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***Abstract*—We propose a sensitivity analysis method that computes the S-parameter Jacobian from the volume E-field solutions in the frequency domain. The field solutions may be provided by any valid electromagnetic analysis. The computation is a post-process, which is independent from the simulator's grid, system equations, and discretization method. The sensitivity solver uses its own finite-difference grid and a sensitivity formula based on the finite-difference frequency-domain vector Helmholtz equation for the electric field. Its computational overhead is negligible in comparison with the simulation. We use the sensitivity solver in the gradient-based optimization of microwave structures. Significant reduction of the time required by the overall optimization process is achieved. In all design examples, the sensitivities are computed from the field solutions provided by a commercial finite-element simulator.**

***Index Terms*—Computer-aided design (CAD), finite-difference method, finite-element method (FEM), gradient-based optimization, response Jacobians, sensitivity analysis.**

I. Introduction

SELF-ADJOINT approaches to the computation of the S-parameter sensitivities have been proposed for use with full-wave high-frequency electromagnetic (EM) simulators, both time and frequency domain [1]–[6]. The response sensitivities are mathematically represented by the response gradient in the space defined by the parameters of interest. In design optimization, such parameters are shape and/or material parameters of particular components of the structure. The sensitivities provide information crucial to design processes such as optimization, tolerance, and yield analyses, as well as modeling (e.g., response surface generation).

The developments in [1]–[6] gain remarkable efficiency from a self-adjoint formulation where the adjoint field solution for each S-parameter is obtained from the original (simulated) $\mathbf{E}$-field solution via straightforward mathematical transformations. In particular, in the frequency domain, the adjoint field is obtained from the $\mathbf{E}$-field solution by a multiplication with a complex constant, the self-adjoint constant [3]. This constant is parameter independent. It depends only on the port excitations. The net result is that the S-parameter Jacobian is computed with very little overhead—it requires less than a second even with hundreds of optimizable parameters. In comparison with full-wave simulation, such overhead is negligible. In addition to being more efficient, the adjoint sensitivity computations are known to be more reliable than the traditional parameter sweeps and response-level finite differences [4], [6], [7].

The main drawback of all previous implementations of the adjoint-variable method, see, e.g., [8] and [9], including the self-adjoint formulations, e.g., [1]–[6], [10], is that they depend on the simulation tool generating the field solution. The main reason is that an adjoint formulation requires the derivatives of the system matrices with respect to the optimizable parameters. The system matrices or the system coefficients describing the EM problem arise from the particular mathematical model and discretization technique adopted by the simulation tool. As pointed out in [3], the system matrices are rarely available with commercial simulation packages. This prevents the implementation of adjoint-variable algorithms unless access is permitted to the core meshing and system-assembly levels of the simulator.

Here, we propose a self-adjoint sensitivity solver for S-parameters, which operates on volume $\mathbf{E}$-field solutions in the frequency domain. Since the EM problem is unique, the field solution can be obtained with any valid analysis technique. It may be obtained also with a time-domain simulation whose output has been transferred into the frequency domain via Fourier transform. The proposed sensitivity solver gains its independence from the simulation method by: 1) generating its own local finite-difference grids fitted to the optimizable objects and 2) using a sensitivity formula based on its own EM model, namely, a finite-difference frequency-domain (FDFD) model. The FDFD discretization is applied to the vector Helmholtz equation for the $\mathbf{E}$-field.

The feasibility of the FDFD paradigm in sensitivity analysis has been investigated in 2-D problems [11] where Yee's TM-mode grid [12] was used. There, Jacobians with respect

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to the shape and material parameters of dielectric objects were considered. Adjoint-based sensitivity analysis of photonic crystal devices based on the 2-D FDFD system equations was reported in [13]. There, the major drawback is the requirement for adjoint analyses, which limits its applicability and degrades its efficiency. The technique also does not allow for the calculation of derivatives with respect to shape parameters. In [14], the use of a central-node finite-difference grid was introduced, which facilitates the implementation of the sensitivity formula in 3-D problems with dielectric and volumetric metallic objects.

Here, we present a general implementation that is capable of handling volumetric dielectric and metallic objects, as well as infinitesimally thin (sheet) metallic objects. For the first time, we also present a complete theory of the self-adjoint sensitivity analysis (SASA) with frequency-domain $\mathbf{E}$-field solutions, which considers both volume excitations (e.g., current density) and port excitations through boundary conditions. Further, we integrate our sensitivity solver in a fully automated optimization loop and demonstrate the time savings realized in gradient-based optimization. The inherent accuracy of the adjoint-based Jacobian computation also leads to improved convergence of the optimization processes.

II. FDFD-Based Sensitivity Formula

A. Response Definitions

The system response can be defined in general as a functional of the complex-phasor field solution $\mathbf{E}$ over the computational volume Ω and/or its boundary surface

$$F(\mathbf{E}) = \iiint_{\Omega} f(\mathbf{E}) d\Omega + \oiint_{S_\Omega} f_s(\mathbf{E}) ds. \tag{1}$$

The response F, as well as the local response functions f and f_s, are scalar and may be complex. The focus here is on the derivative of F with respect to an optimizable parameter p_n ($n = 1, \ldots, N$). Using the chain rule, (1) is differentiated as

$$\frac{\partial F}{\partial p_n} = \frac{\partial^e F}{\partial p_n} + \iiint_{\Omega} \left(\sum_{\xi=x,y,z} \frac{\partial f}{\partial E_\xi} \cdot \frac{\partial E_\xi}{\partial p_n} \right) d\Omega + \oiint_{S_\Omega} \left(\sum_{\xi=x,y,z} \frac{\partial f_s}{\partial E_\xi} \cdot \frac{\partial E_\xi}{\partial p_n} \right) ds. \tag{2}$$

The first term in (2) is the explicit derivative of F with respect to p_n. The integral terms reflect the implicit dependence of F on p_n through the $\mathbf{E}$-field solution.

B. FDFD System Equations

Consider the $\mathbf{E}$-field vector Helmholtz equation,

$$\nabla \times \boldsymbol{\mu}_r^{-1} \nabla \times \mathbf{E} - k_0^2 \boldsymbol{\varepsilon}_r \mathbf{E} + j\omega\mu_0 \boldsymbol{\sigma} \mathbf{E} = -j\omega\mu_0 \mathbf{J}. \tag{3}$$

Here, $\boldsymbol{\mu}_r$, $\boldsymbol{\varepsilon}_r$, and $\boldsymbol{\sigma}$ are the relative permeability, permittivity, and conductivity. These are, in general, complex tensors. For simplicity, here we focus on isotropic media, although the adjoint-variable approach is, in general, applicable to anisotropic media. The wavenumber $k_0 = \omega\sqrt{\mu_0\varepsilon_0}$ is that in vacuum with μ_0 and ε_0 being the permeability and permittivity of vacuum. The current density is given by $\mathbf{J}$.

Central finite differences are applied to discretize (3) on a central-node rectangular hexahedral grid [14], which results in the difference equation

$$C^2\mathbf{E} + \alpha\mathbf{E} = \mathbf{G} \tag{4}$$

where C^2 is a representation of the double-curl operator

$$C^2 \approx -\Delta h^2 \nabla \times \mu_r^{-1} \nabla \tag{5}$$

with $\Delta h = \min(\Delta x, \Delta y, \Delta z)$ being the smallest discretization step in space. The mesh allows for different discretization steps, Δx, Δy, and Δz along the three axes. The numerical constant α is

$$\alpha = (k_0 \Delta h)^2 \left[\varepsilon_r - j\varepsilon_r \tan\delta_d - j\sigma(\omega\varepsilon_0)^{-1} \right] \tag{6}$$

whose imaginary part reflects conduction loss (through σ) and dielectric polarization loss (through $\tan\delta_d$). The right-hand term $\mathbf{G}$ in (4) reflects the presence of current density. In the context of sensitivity analysis, it is important to identify two contributions to this term. First, if impressed current density $\mathbf{J}^i$ is present, then $\mathbf{G} = \mathbf{G}^i = \beta\mathbf{J}^i$, $\beta = j\omega\mu_0\Delta h^2$. Second, in the case of perfect electric conductor (PEC) objects, whose shapes are optimizable, it becomes necessary to consider the induced (or secondary) current density at their surface [15]. The secondary-current term is $\mathbf{G} = \mathbf{G}^s = C^2\mathbf{E}$ since $\alpha\mathbf{E} = 0$ at the center of the PEC cell in a central-node finite-difference grid [16], [17]. $\mathbf{G}^s$ is computed from the $\mathbf{E}$-field solution in the immediate vicinity of the metallic object and is thus nonzero only at the cells forming the object's surface layer. Note that, at a given grid point, the two terms $\mathbf{G}^i$ and $\mathbf{G}^s$ are never simultaneously nonzero.

The implementation of the C^2 operator on a central-node finite-difference grid is illustrated below by its x component

$$(C^2\mathbf{E})_x = h_y^2 D_{yy}^{\mu} E_x + h_z^2 D_{zz}^{\mu} E_x - h_y h_x D_{yx}^{\mu} E_y - h_z h_x D_{zx}^{\mu} E_z \tag{7}$$

where $h_\xi = \Delta h/\Delta\xi$, $\xi \equiv x, y, z$. The second-order operators D_{yy}^{μ} and D_{yx}^{μ} at the grid point (x_0, y_0, z_0) are

$$\begin{aligned}(D_{yy}^{\mu} E_x)_{(x_0,y_0,z_0)} &= \frac{E_{x(x_0,y_0+\Delta y,z_0)}}{\mu_{r(x_0,y_0+\Delta y/2,z_0)}} + \frac{E_{x(x_0,y_0-\Delta y,z_0)}}{\mu_{r(x_0,y_0-\Delta y/2,z_0)}} \\ &\quad - \left[\frac{1}{\mu_{r(x_0,y_0+\Delta y/2,z_0)}} + \frac{1}{\mu_{r(x_0,y_0-\Delta y/2,z_0)}} \right] \cdot E_{x(x_0,y_0,z_0)} \end{aligned} \tag{8}$$

$$\begin{aligned}(D_{yx}^{\mu} E_y)_{(x_0,y_0,z_0)} &= \frac{1}{4} \left[\frac{E_{y(x_0+\Delta x,y_0+\Delta y,z_0)} - E_{y(x_0-\Delta x,y_0+\Delta y,z_0)}}{\mu_{r(x_0,y_0+\Delta y,z_0)}} \right. \\ &\quad \left. - \frac{E_{y(x_0+\Delta x,y_0-\Delta y,z_0)} - E_{y(x_0-\Delta x,y_0-\Delta y,z_0)}}{\mu_{r(x_0,y_0-\Delta y,z_0)}} \right]. \end{aligned} \tag{9}$$

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TABLE I
C_x^2 COEFFICIENTS AT THE POINT (i,j,k), $\mu_r = 1$

Grid position	E_x	E_y	E_z
i,j,k	$-2(h_y^2+h_z^2)$	0	0
$i-1,j,k$	0	0	0
$i+1,j,k$	0	0	0
$i,j-1,k$	h_y^2	0	0
$i,j+1,k$	h_y^2	0	0
$i,j,k-1$	h_z^2	0	0
$i,j,k+1$	h_z^2	0	0
$i-1,j-1,k$	0	$-h_xh_y/4$	0
$i+1,j-1,k$	0	$h_xh_y/4$	0
$i-1,j+1,k$	0	$h_xh_y/4$ *	0
$i+1,j+1,k$	0	$-h_xh_y/4$	0
$i-1,j,k-1$	0	0	$-h_xh_z/4$
$i+1,j,k-1$	0	0	$h_xh_z/4$
$i-1,j,k+1$	0	0	$h_xh_z/4$
$i+1,j,k+1$	0	0	$-h_xh_z/4$
$i,j-1,k-1$	0	0	0
$i,j+1,k-1$	0	0	0
$i,j-1,k+1$	0	0	0
$i,j+1,k+1$	0	0	0

The operators D^{μ}_{zz} and D^{μ}_{zx} are analogous to D^{μ}_{yy} and D^{μ}_{yx}, respectively, with the sampling being along z instead of y. As an example, the C_x^2 system coefficients producing $(C^2\mathbf{E})_x$ at the point (i,j,k) are summarized in Table I for the case of $\mu_r = 1$. The y and z components of $C^2\mathbf{E}$ can be obtained from $(C^2\mathbf{E})_x$ by the cyclic substitution $x \to y \to z \to x$.

C. Adjoint Field Definition and Sensitivity Formula

Assume that (3) has been solved for the nominal (unperturbed) structure and the respective field solution $\bar{\mathbf{E}}$ has been found in Ω and at S_Ω. $\bar{\mathbf{E}}$ then satisfies the equation

$$C_0^2\bar{\mathbf{E}} + \alpha_0\bar{\mathbf{E}} = \mathbf{G}_0 \tag{10}$$

where subscript 0 emphasizes that the system coefficients reflect the materials and excitation in the nominal state.

A perturbation Δp_n in the nth parameter p_n changes the local material parameters due to shape or material changes of the affected object. Thus, the local equation coefficients (C^2, α, G_ξ, $\xi = x,y,z$) are affected, which leads to a change in the field solution $\Delta_n\mathbf{E}$. The nth perturbed state, where the nth parameter is $p_n + \Delta p_n$, is described by

$$(C_0^2 + \Delta_n C^2)(\bar{\mathbf{E}} + \Delta_n\mathbf{E}) + (\alpha_0 + \Delta_n\alpha)(\bar{\mathbf{E}} + \Delta_n\mathbf{E}) = (\mathbf{G}_0 + \Delta_n\mathbf{G}) \tag{11}$$

where Δ_n denotes a change due to the perturbation of p_n. $\Delta_n\mathbf{E} = \bar{\mathbf{E}}_n - \bar{\mathbf{E}}$, where $\bar{\mathbf{E}}_n$ is also the solution in the nth perturbed state. Taking into account (10), (11) reduces to

$$C_n^2\Delta_n\mathbf{E} + \alpha_n\Delta_n\mathbf{E} + \Delta_n C^2\bar{\mathbf{E}} + \Delta_n\alpha\bar{\mathbf{E}} - \Delta_n\mathbf{G} = 0 \tag{12}$$

where $C_n^2 = C_0^2 + \Delta_n C^2$ and $\alpha_n = \alpha_0 + \Delta_n\alpha$ are the local equation coefficients corresponding to the nth perturbed state.

Taking a dot product of both sides of (12) with a field $\hat{\mathbf{E}}_n$ (for now arbitrary) and integrating over Ω results in

$$\iiint_\Omega \hat{\mathbf{E}}_n \cdot \big(C_n^2\Delta_n\mathbf{E} + \alpha_n\Delta_n\mathbf{E} + \Delta_n C^2\bar{\mathbf{E}} + \Delta_n\alpha\bar{\mathbf{E}} - \Delta_n\mathbf{G}\big)d\Omega = 0. \tag{13}$$

This expression allows to relate the derivative (2) to the field solution $\bar{\mathbf{E}}$. Bearing in mind that C_n^2 is a finite-difference representation of the double-curl operator [see (5)] in the nth perturbed structure, the first term in (13) is transformed as[1]

$$\begin{aligned}\iiint_\Omega \hat{\mathbf{E}}_n \cdot (C_n^2\Delta_n\mathbf{E})d\Omega &\approx \iiint_\Omega \Delta_n\mathbf{E} \cdot (C_n^2\hat{\mathbf{E}}_n)d\Omega \\ &\quad - \Delta h^2 \oiint_{S_\Omega} \big[\Delta_n\mathbf{E} \times (\mu_r^{-1}\nabla \times \hat{\mathbf{E}}_n) \\ &\quad - \hat{\mathbf{E}}_n \times (\mu_r^{-1}\nabla \times \Delta_n\mathbf{E})\big] \cdot d\mathbf{s}.\end{aligned} \tag{14}$$

The expression in (14) is substituted into (13) and the result is divided by Δp_n. Thus, the derivative $\partial\mathbf{E}/\partial p_n \approx \Delta_n\mathbf{E}/\Delta p_n$ is introduced, which now fulfills the equation

$$\iiint_\Omega \frac{\partial\mathbf{E}}{\partial p_n} \cdot (C_n^2\hat{\mathbf{E}}_n + \alpha_n\hat{\mathbf{E}}_n)d\Omega - \Delta h^2 I_S \approx -\iiint_\Omega \hat{\mathbf{E}}_n \cdot \frac{\Delta_n R(\bar{\mathbf{E}})}{\Delta p_n}d\Omega \tag{15}$$

where

$$I_S = \oiint_{S_\Omega} \left[\frac{\partial\mathbf{E}}{\partial p_n} \times (\mu_r^{-1}\nabla \times \hat{\mathbf{E}}_n) - \hat{\mathbf{E}}_n \times \left(\mu_r^{-1}\nabla \times \frac{\partial\mathbf{E}}{\partial p_n}\right)\right] \cdot d\mathbf{s} \tag{16}$$

and

$$\frac{\Delta_n R(\bar{\mathbf{E}})}{\Delta p_n} = \frac{\Delta_n C^2\bar{\mathbf{E}}}{\Delta p_n} + \frac{\Delta_n\alpha}{\Delta p_n}\bar{\mathbf{E}} - \frac{\Delta_n\mathbf{G}}{\Delta p_n}. \tag{17}$$

Notice that all terms in (17) are known and their calculation requires only the field solution $\bar{\mathbf{E}}$ at grid points where the system coefficients are affected by the perturbation Δp_n.

1) Adjoint Field: We aim at making the left-hand side of (15) equivalent to the integral terms in (2).

[1]The integral vector identity used in this transformation is

$$\iiint_\Omega (\mathbf{B} \cdot \nabla \times \mu_r^{-1}\nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mu_r^{-1}\nabla \times \mathbf{B})d\Omega = \oiint_{S_\Omega} (\mathbf{A} \times \mu_r^{-1}\nabla \times \mathbf{B} - \mathbf{B} \times \mu_r^{-1}\nabla \times \mathbf{A}) \cdot d\mathbf{s}.$$

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Comparing the left volume integral in (15) with that in (2), we see that the two become equivalent if $\hat{\mathbf{E}}_n$ is constrained to satisfy the system equation

$$C_n^2\hat{\mathbf{E}}_n + \alpha_n\hat{\mathbf{E}}_n = \hat{\mathbf{G}}_n \tag{18}$$

where $\hat{\mathbf{G}}_n$ reflects impressed adjoint excitation $\hat{\mathbf{G}}^i$ given by

$$\hat{\mathbf{G}}_n = \hat{\mathbf{G}}^i = \sum_{\xi=x,y,z} \mathbf{a}_\xi \left.\frac{\partial f}{\partial E_\xi}\right|_{E_\xi=\bar{E}_\xi} \tag{19}$$

Here, $\mathbf{a}_\xi$ is a coordinate unit vector. Notice that the impressed-current adjoint excitation $\hat{\mathbf{G}}^i$ is parameter-independent; hence, the lack of the subscript n. However, the secondary-current term (at the surface of optimizable PEC objects) $\hat{\mathbf{G}}_n = \hat{\mathbf{G}}_n^s = C_n^2\hat{\mathbf{E}}_n$ is parameter dependent. Note also that $\hat{\mathbf{E}}_n$ satisfies the system equation corresponding to the nth perturbed state—the same as that for the original field $\bar{\mathbf{E}}_n$. $\hat{\mathbf{E}}_n$ is the adjoint field [8] in the nth perturbed state.[2] Similar observations were made for implementations based on the finite-element method (FEM) [3] and the finite-difference time-domain method [4], [15].

The next step is to make the surface integral I_S in (15) and (16) equivalent to the surface integral in (2). This leads to additional restrictions on the adjoint field $\hat{\mathbf{E}}_n$, which define its boundary conditions at S_Ω. Here, only some simple and relevant cases are considered.

Assume first that $f_s = 0$ on $S_{\Omega 0} \subseteq S_\Omega$, i.e., the response of interest does not depend on the field solution at $S_{\Omega 0}$. $I_{S0} = 0$ must then be fulfilled where I_{S0} is the portion of (16) taken over $S_{\Omega 0}$. This is achieved by requiring that the adjoint field $\hat{\mathbf{E}}_n$ satisfies the same boundary conditions as $\bar{\mathbf{E}}$ and $\bar{\mathbf{E}}_n$ at $S_{\Omega 0}$. Note that (16) depends on the boundary conditions of $\partial\mathbf{E}/\partial p_n$, but these are identical with the boundary conditions of $\bar{\mathbf{E}}$ and $\bar{\mathbf{E}}_n$. In general, $S_{\Omega 0}$ may consist of PEC, perfect magnetic conductor (PMC), and radiation (or reflection-free) boundary conditions (RBCs).[3] With $\boldsymbol{\mathcal{E}} \equiv \hat{\mathbf{E}}$, $\partial\mathbf{E}/\partial p_n$, the boundary conditions are $\mathbf{a}_n \times \boldsymbol{\mathcal{E}} = 0$ at PEC, $\mathbf{a}_n \times (\nabla \times \boldsymbol{\mathcal{E}}) = 0$ at PMC, and $\nabla \times \boldsymbol{\mathcal{E}} = -\gamma(\mathbf{a}_n \times \boldsymbol{\mathcal{E}})$ at RBC. Here, $\mathbf{a}_n$ denotes the outward unit normal of the integration surface, i.e., $d\mathbf{s} = \mathbf{a}_n ds$, and γ is the propagation constant. That the above boundary conditions result in $I_{S0} = 0$ is proven using triple-product vector identities.

Next, we focus on S-parameter analysis where the port cross sections are parts of S_Ω. For K ports, the derivatives of all S_{kj} parameters $(j,k = 1,\ldots,K)$ may be of interest. When considering the S_{kj} derivatives, two factors are important, which are: 1) the excitation can be done through an inhomogeneous boundary condition at the jth-port (the remaining ports are matched, i.e., the respective homogeneous boundary condition is employed) and 2) the response depends only on the field solution (the outgoing wave) at the surface S_k of the kth-port, i.e., f_s is zero everywhere on S_Ω, except S_k.

When the jth-port is excited, the boundary condition at the ith-port can be written in a general form as [19], [20]

$$\mathbf{a}_{n,i} \times (\nabla \times \mathbf{E}) + P(\mathbf{E}) = \mathbf{U}_i^{\text{inc}}, \qquad i = 1,\ldots,K \tag{20}$$

$$\mathbf{U}_i^{\text{inc}} = \begin{cases} 2P(\mathbf{E}_{i\tan}^{\text{inc}}), & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} \tag{21}$$

where $\mathbf{E}_{j\tan}^{\text{inc}}$ is the incident field tangential to the port surface. The operator P is a linear vector operator, which depends on the adopted boundary condition. In the simplest case of a first-order boundary condition, $P(\mathbf{E}) = \gamma_i \mathbf{a}_{n,i} \times (\mathbf{a}_{n,i} \times \mathbf{E})$. Here, γ_i is the propagation constant of the mode of interest at the ith-port and $\mathbf{a}_{n,i}$ is the port outward unit normal.

Since I_S vanishes on the PEC, PMC, and RBC portions of S_Ω, it reduces to an integral over the port cross sections

$$I_S = \sum_{i=1}^{K} \iint_{S_i} \left[\frac{\partial\mathbf{E}}{\partial p_n} \times (\mu_r^{-1}\nabla \times \hat{\mathbf{E}}_n) - \hat{\mathbf{E}}_n \times \left(\mu_r^{-1}\nabla \times \frac{\partial\mathbf{E}}{\partial p_n}\right)\right] \cdot d\mathbf{s} \tag{22}$$

Similarly to the other boundary surfaces, we require that $\hat{\mathbf{E}}_n$ satisfies the same condition as in (20) with excitations $\hat{\mathbf{U}}_i^{\text{inc}}$, $i = 1,\ldots,K$ yet to be determined. Making use of (20) and the triple-product identities, (22) can now be written as

$$I_S = \sum_{i=1}^{K} \iint_{S_i} \mu_r^{-1} \left\{ \frac{\partial\mathbf{E}_{\tan}}{\partial p_n} \cdot \left[P(\hat{\mathbf{E}}_n) - \hat{\mathbf{U}}_i^{\text{inc}}\right] - (\hat{\mathbf{E}}_n)_{\tan} \cdot \left[P\left(\frac{\partial\mathbf{E}}{\partial p_n}\right) - \frac{\partial\mathbf{U}_i^{\text{inc}}}{\partial p_n}\right]\right\} ds \tag{23}$$

where $\mathbf{E}_{\tan}$ and $(\hat{\mathbf{E}}_n)_{\tan}$ are the tangential components of the respective fields at the port. Note that when $i \neq j$, $\mathbf{U}_i^{\text{inc}} = 0$ and, therefore, $\partial\mathbf{U}_i^{\text{inc}}/\partial p_n = 0$. For $i = j$, $\mathbf{U}_i^{\text{inc}} = 2P(\mathbf{E}_{i\tan}^{\text{inc}})$. Provided that the jth-port is not affected by the perturbation in p_n, the condition $\partial\mathbf{U}_i^{\text{inc}}/\partial p_n = 0$ still holds. Also, in the Appendix, it is shown that for all $S_i (i = 1,\ldots,K)$

$$\iint_{S_i} \mu_r^{-1} \frac{\partial\mathbf{E}_{\tan}}{\partial p_n} \cdot P(\hat{\mathbf{E}}_n) ds = \iint_{S_i} \mu_r^{-1} (\hat{\mathbf{E}}_n)_{\tan} \cdot P\left(\frac{\partial\mathbf{E}}{\partial p_n}\right) ds. \tag{24}$$

Finally, (23) yields

$$I_S = -\sum_{i=1}^{K} \iint_{S_i} \mu_r^{-1} \frac{\partial\mathbf{E}}{\partial p_n} \cdot \hat{\mathbf{U}}_i^{\text{inc}} ds. \tag{25}$$

I_S from (25) is substituted in (15), where the term $-\Delta h^2 I_S$ can now be made equal to the surface integral in (2). Bearing in

[2]For anisotropic media, the adjoint system requires the use of the *transposed* constitutive tensors [15]. If these are symmetric, the adjoint equations remain the same as the original ones. A word of caution is due regarding the term "adjoint" describing (18) and its solution $\hat{\mathbf{E}}_n$. Transposed constitutive tensors imply that the respective system operator $\mathbb{L}^T$ is the "transpose" of the EM operator $\mathbb{L}$ so that $\langle f, \mathbb{L}g\rangle = \langle g, \mathbb{L}^T f\rangle$ holds [18]. This is different from the mathematical definition of the operator $\mathbb{L}^a$ adjoint to $\mathbb{L}$, $\langle f^*, \mathbb{L}g\rangle = \langle g^*, \mathbb{L}^a f\rangle^*$. Upon discretization, $\mathbb{L}^a$ leads to system matrices, which are transpose conjugate of those produced by $\mathbb{L}$. Since $\hat{\mathbf{E}}_n$ solves a discretized version of $\mathbb{L}^T$ (not $\mathbb{L}^a$), conjugation is *not* applied regardless of whether the medium is anisotropic and/or lossy. Strictly speaking, $\hat{\mathbf{E}}_n$ should be referred to as the "pseudoadjoint" solution although here we chose to adhere to the traditional term "adjoint solution" [8]. Detailed discussion on the transpose and adjoint of an operator, as well as the respective inner products used in electromagnetism, can be found in [18].

[3]Other boundary conditions, e.g., impedance boundary conditions, can be handled similarly.

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mind that f_s is nonzero only at the kth-port, the adjoint excitation is obtained as

$$\hat{\mathbf{U}}_i^{\text{inc}} = \begin{cases} \dfrac{\mu_r}{\Delta h^2}\left(\dfrac{\partial f_s}{\partial E_{t1}}\mathbf{a}_{t1} + \dfrac{\partial f_s}{\partial E_{t2}}\mathbf{a}_{t2}\right)_{E_\xi = \bar{E}_\xi}, & \text{if } i = k \\ 0, \quad \text{if } i \neq k. \end{cases} \tag{26}$$

Here, $\mathbf{a}_{t1}$ and $\mathbf{a}_{t2}$ are the tangential unit vectors at the port and the tangential field at the port is $\mathbf{E}_{\tan} = E_{t1}\mathbf{a}_{t1} + E_{t2}\mathbf{a}_{t2}$. The exact forms of f_s and $\hat{\mathbf{U}}_k^{\text{inc}}$ at the kth-port are discussed in the following section.

In summary, the adjoint field is uniquely defined by its governing (18), its sources (19), and its boundary conditions. The latter are the same as those of the original field $\bar{\mathbf{E}}$, except for the adjoint boundary excitation (26), which is nonzero at the kth-port since the response depends on the outgoing field there. In contrast, the original problem is excited at the jth-port as per (20) and (21).

2) Sensitivity Formula: Through the constraints imposed on $\hat{\mathbf{E}}_n$, (15) is transformed into the sensitivity formula

$$\frac{\partial F}{\partial p_n} \approx \frac{\partial^e F}{\partial p_n} - \iiint_\Omega \hat{\mathbf{E}}_n \cdot \frac{\Delta_n R(\bar{\mathbf{E}})}{\Delta p_n} d\Omega \tag{27}$$

where the difference residual term $\Delta_n R(\bar{\mathbf{E}})/\Delta p_n$ has been defined in (17). The nth adjoint field $\hat{\mathbf{E}}_n$ is found by solving (18) with its respective sources and boundary conditions.

The sensitivity formula (27) is second order in the sense that it is derived from the perturbation expression (11) where the second-order terms $\Delta_n\alpha \cdot \Delta_n\mathbf{E}$ and $\Delta_n C^2(\Delta_n\mathbf{E})$ are retained. Its advantage is that the difference ratios of the system coefficients in (17) do not have to be good representations of the respective derivatives. This is important when the derivatives of the system coefficients are not available analytically, e.g., in the case of shape parameters of structures discretized on a finite-difference grid [4], [15]. Its disadvantage is that the adjoint field is parameter dependent, which is emphasized by the subscript n in $\hat{\mathbf{E}}_n$ and its equation (18). This implies the need to perform as many adjoint system analyses as the number of optimizable parameters N. In practice, this disadvantage is easily overcome using the mapping of Bakr *et al.* [21]. With this mapping, only the adjoint field $\hat{\mathbf{E}}$ of the nominal structure is found by solving

$$C_0^2\hat{\mathbf{E}} + \alpha_0\hat{\mathbf{E}} = \hat{\mathbf{G}}_0 \tag{28}$$

where $\mathbf{G}_0 = \hat{\mathbf{G}}^i$ is the same as in (19). The boundary conditions complementing (28) are the same as those of $\hat{\mathbf{E}}_n$. All parameter-dependent adjoint fields $\hat{\mathbf{E}}_n$, $n = 1, \ldots, N$ are obtained from $\hat{\mathbf{E}}$ by a coordinate shift in the direction of the assumed parameter perturbation Δp_n. For example, if $\Delta p_n = \Delta x$, then $\hat{\mathbf{E}}_n(x + \Delta x, y, z) = \hat{\mathbf{E}}(x, y, z)$ at the affected object boundary. Note that the adjoint fields are needed only at grid points in the immediate vicinity of optimizable shapes. The mapping has no computational overhead.

If the second-order terms in (11) are neglected, the analytic (first-order) sensitivity formula is derived as follows:

$$\frac{\partial F}{\partial p_n} = \frac{\partial^e F}{\partial p_n} - \iiint_\Omega \hat{\mathbf{E}} \cdot \frac{\partial R(\bar{\mathbf{E}})}{\partial p_n} d\Omega \tag{29}$$

where

$$\frac{\partial R(\bar{\mathbf{E}})}{\partial p_n} = \frac{\partial C^2\bar{\mathbf{E}}}{\partial p_n} + \frac{\partial \alpha}{\partial p_n}\bar{\mathbf{E}} - \frac{\partial \mathbf{G}}{\partial p_n}. \tag{30}$$

The derivation follows the same steps as for the second-order formula. In (30), the analytical derivatives of the system coefficients (or their sufficiently accurate approximations) are required. The adjoint field $\hat{\mathbf{E}}$ is parameter independent and it is the solution of (28).

III. Implementation on a Central-Node Finite-Difference Grid

A. Shape Parameters of Metallic Objects

Metallic (or PEC) shape parameters in the general 3-D case are treated using the same principles as in the 2-D case [4], [15]. The assumed perturbation Δp_n is chosen to be equal to one grid step size along the respective axis. As an example, a cross-sectional view of a metallic septum is shown in Fig. 1 on a finite-difference grid whose step size equals the septum thickness. In Fig. 1(a), the assumed perturbation of the length p_n of the metallic septum is $\Delta p_n = \Delta x$. The septum in the nominal state is shown in dark gray. The septum in the nth perturbed state is longer and includes the light-gray cell. The light-gray cell is thus "metallized" as a result of the perturbation. While in the nominal state, the field components in this cell are not zero, in the nth perturbed state, they must be set to zero. This change affects the C^2 system coefficients at all neighboring points [marked with black circles in Fig. 1(a)]. As an illustration, consider the point (i, j, k) in Fig. 1(a). Its C_x^2 coefficient for the location $(i-1, j+1, k)$ changes from its nominal value (marked with a star in Table I) to zero, i.e., $\Delta_n C_x^2 = -h_x h_y/4$. The rest of its C_x^2 coefficients do not change. Since $\bar{E}_{y(i-1,j+1,k)}$ is not zero, the term $(\Delta_n C^2\bar{\mathbf{E}})_x = -(h_x h_y/4) \cdot \bar{E}_{y(i-1,j+1,k)}$ at (i, j, k) contributes to the residual difference term $\Delta_n R(\bar{\mathbf{E}})$ in (17). In (27), this term is multiplied by a nonzero $\hat{E}_{n,x(i,j,k)}$ and contributes to the overall sensitivity value. The procedure is repeated for all points whose double-curl operator is affected by the cell metallization, and the contributions are added up as per the integration in (27).

Note that the C^2 operator is affected also at the "metallized" point $(i-1, j+1, k)$ and so is the coefficient α, which changes from its value in (6) to zero. These changes in the system coefficients result in a nonzero $\Delta_n R(\bar{\mathbf{E}})$ term at $(i-1, j+1, k)$; however, this term has no consequence because in the integrand of (27) it is multiplied by $\hat{\mathbf{E}}_{n(i-1,j+1,k)} = 0$.

Assumed shape perturbations may also involve "demetallization" of cells, which is illustrated in Fig. 1(b) by the perturbation $\Delta p_n = -\Delta x$. Similarly to the 2-D case with metallic objects, the 3-D analysis [16] shows that the $(\Delta_n C^2\bar{\mathbf{E}})$ term is always zero due to zero values of $\bar{\mathbf{E}}$ at the "demetallized" cells [light-gray cell at $(i-2, j+1, k)$ in Fig. 1(b)]. However, changes are invoked in the induced current densities at the metallic surfaces, which result in nonzero $\Delta_n\mathbf{G}$ terms [see (17)]. In Fig. 1(b), for example, $\Delta_n\mathbf{G}$ is nonzero only at $(i-2, j+1, k)$ because $\mathbf{G}_0^s$ is nonzero in the nominal state ($\mathbf{G}_0^s = C_0^2\bar{\mathbf{E}}$, light-gray cell is PEC), while it is zero in the nth perturbed state ($\mathbf{G}_n^s = 0$, light-gray cell is dielectric). Thus, $\Delta_n\mathbf{G} =$

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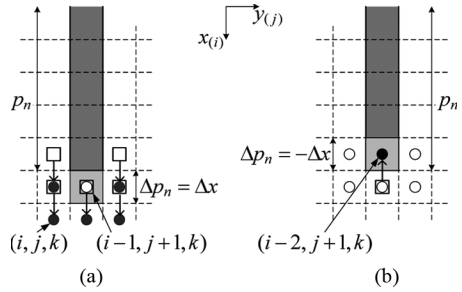


Fig. 1. Illustration of assumed shape perturbations of a metallic object. (a) Increase in the length p_n by Δx. (b) Decrease in p_n by Δx. The light-gray cell indicates the location where the medium changes from: (a) dielectric to PEC and (b) from PEC to dielectric. White circles indicate where $\bar{\mathbf{E}}$ is recorded. Black circles indicate where $\hat{\mathbf{E}}_n$ is needed. Squares indicate where the adjoint field $\hat{\mathbf{E}}$ is recorded in the nominal state and the short arrows indicate the direction of the mapping (coordinate shift) used to obtain $\hat{\mathbf{E}}_n$ from $\hat{\mathbf{E}}$.

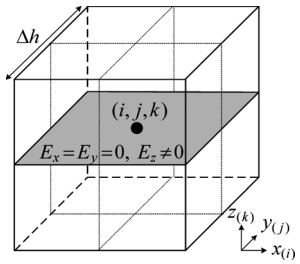


Fig. 2. Illustration of a central-node cubical cell of size Δh and the location of a metallic infinitesimally thin patch in the x–y-plane.

$-C_0^2\bar{\mathbf{E}}$ and it is calculated from $\bar{\mathbf{E}}$ at the points marked with white circles. Also, $\hat{\mathbf{E}}_{n(i-2,j+1,k)}$ is not zero leading to a contribution to the sensitivity integral in the form of $-(C_0^2\bar{\mathbf{E}})\cdot\hat{\mathbf{E}}_n$ calculated at $(i-2, j+1, k)$.

Note that, in general, the field $\bar{\mathbf{E}}$ and its adjoint counterpart $\hat{\mathbf{E}}$ are needed in a limited number of cells in the immediate vicinity of the surface that changes as a result of an assumed perturbation. For example, in Fig. 1, the locations at which $\bar{\mathbf{E}}$ is recorded are denoted with white circles, while those for $\hat{\mathbf{E}}$ are shown by the squares. Thus, the memory and time requirements of calculating the sensitivity via (27) are negligible. Note also that with the central-node grid, the three $\mathbf{E}$-field components are never recorded on the faces, edges, or vertices of the metallic boundary. For example, for a cubical cell of size Δh, the nodes where the field is recorded are at distances of $\Delta h/2$, $\Delta h/\sqrt{2}$, and $\Delta h\sqrt{3}/2$ from adjacent metallic boundary faces, edges, or vertices, respectively.

In addition, one may consider infinitesimally thin metallic objects. The location of a metallic-sheet patch on a central-node grid is illustrated in Fig. 2. This time, a node rests in the middle of a patch where the tangential $\mathbf{E}$-field components vanish while the normal component does not. The assumed metallization or demetallization of surface patches is handled similarly to the volumetric case. The difference is that the system coefficients multiplying the normal field components are not affected by the assumed perturbations [17]. As in the volumetric case, mesh nodes are always away from the metallic edges and vertices. For a cubical cell, the distance to an edge is $\Delta h/2$, while the distance to a vertex is $\Delta h/\sqrt{2}$.

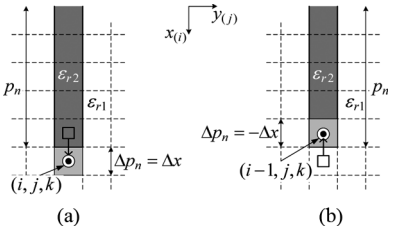


Fig. 3. Illustration of assumed shape perturbations of a dielectric object. (a) Increase in the length p_n by Δx. (b) Decrease in p_n by Δx. The light-gray cell indicates the location where the medium changes from: (a) dielectric 1 to dielectric 2 and (b) from dielectric 2 to dielectric 1. $\bar{\mathbf{E}}$ and $\hat{\mathbf{E}}_n$ are needed at the same locations (white and black circles overlap). The nominal-state adjoint field $\hat{\mathbf{E}}$ is recorded at the locations indicated by squares and the mapping in obtaining $\hat{\mathbf{E}}_n$ is indicated by short arrows.

In summary, an assumed shape perturbation of a metallic object contributes to the sensitivity integral (27) through the terms $\Delta_n C^2\bar{\mathbf{E}}$ (metallization) and $\Delta_n\mathbf{G}$ (demetallization).

B. Shape Parameters of Dielectric Objects

Assumed perturbations of the shape of a dielectric object affect only the system coefficient α at cells whose permittivity changes as a result of the perturbation. This is illustrated in Fig. 3, which shows a cross-sectional view of a dielectric object of thickness of one grid cell. The surrounding medium is of relative permittivity ε_{r1} while that of the object is ε_{r2}. Both are loss free. An assumed increase in the length $\Delta p_n = \Delta x$ [see Fig. 3(a)] would result in a change of $\Delta_n\alpha = k_0\Delta h^2(\varepsilon_{r2}-\varepsilon_{r1})$ at the point (i, j, k) leading to a contribution in the sensitivity integral as $(\Delta_n\alpha/\Delta x)(\hat{\mathbf{E}}_n\cdot\bar{\mathbf{E}})$ where both $\hat{\mathbf{E}}_n$ and $\bar{\mathbf{E}}$ are taken at (i, j, k). Similarly, an assumed decrease in the length $\Delta p_n = -\Delta x$ [see Fig. 3(b)] results in $\Delta_n\alpha = k_0\Delta h^2(\varepsilon_{r1}-\varepsilon_{r2})$ at the location $(i-1, j, k)$ and a respective contribution to the sensitivity integral. In fact, in the simple example of Fig. 3, the above-mentioned points are the only locations where the integrand of (27) is nonzero.

Notice that assumed shape perturbations of dielectric objects contribute to the sensitivity integral (27) through the term $\Delta_n\alpha\cdot\bar{\mathbf{E}}$ only. This is also the case if conductivity is involved.

C. Material Parameters

The system coefficients of the FDFD equation (4) are analytical functions of the material parameters and are thus differentiable. It is evident from (5) and (6) that the C^2 coefficients depend on the relative permeability, while α depends on the permittivity and conductivity. Thus, when a constitutive parameter of an object is optimizable, the exact sensitivity formula (29) and (30) is used. There, typically $\partial\mathbf{G}/\partial p_n = 0$, i.e., it is assumed that the current density source does not depend on the material parameters. Also,

$$\frac{\partial\alpha}{\partial p_n} = \begin{cases} (k_0\Delta h)^2(1-j\tan\delta_d), & \text{if } p_n = \varepsilon_r \\ -j\frac{(k_0\Delta h)^2}{\omega\varepsilon_0}, & \text{if } p_n = \sigma \\ -j\varepsilon_r(k_0\Delta h)^2, & \text{if } p_n = \tan\delta_d \end{cases} \tag{31}$$

and

$$\frac{\partial C^2}{\partial\mu_r} = -\frac{C^2}{\mu_r}. \tag{32}$$

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In order to compute the derivative with respect to a material parameter of an object, the integration in (29) is performed over the object's volume where $\bar{\mathbf{E}}$ and $\hat{\mathbf{E}}$ must be recorded. In comparison with the shape parameters, this involves a larger number of cells where the field values are stored and processed. Thus, material sensitivities may require more computational resources than the shape sensitivities. Yet these requirements are significantly less than those of the full-wave simulation. In the unlikely scenario of the material parameters being optimizable in the entire computational domain, the memory requirement becomes equivalent to that of the simulation. However, the time requirement remains insignificant since the sensitivity computation is a simple volume integration, as per (29)–(30). The sensitivity analysis is a post-process, which is also carried out after the simulation, and thus, does not impede it in any way.

IV. S-PARAMETER SENSITIVITIES

The S-parameter sensitivity formula is derived using the S-parameter formulation [20]

$$S_{kj}^{(\upsilon)} = \frac{\iint_{S_k} \left(\mathbf{E}_j \times \mathbf{h}_k^{(\upsilon)}\right) \cdot d\mathbf{s}}{\iint_{S_j} \left(\mathbf{E}_j^{\text{inc}} \times \mathbf{h}_j^{(\upsilon)}\right) \cdot d\mathbf{s}} - \delta_{kj}$$

$$\delta_{kj} = \begin{cases} 1, & k = j \\ 0, & k \neq j. \end{cases} \tag{33}$$

where υ denotes the desired mode, $\mathbf{h}_\zeta^{(\upsilon)}, \zeta = j, k$ is the dual (or magnetic) port modal vector, $\mathbf{E}_j$ is the field resulting from exciting the jth-port, and $\mathbf{E}_j^{\text{inc}}$ is the incident field at the jth-port. The port surfaces are S_k and S_j, respectively. Only the $\mathbf{E}$-field components that are tangential to the port surfaces contribute to the flux integrals, i.e., the field of interest at S_k is $\mathbf{E}_{j\tan}$, while at S_j, it is $\mathbf{E}_{j\tan}^{\text{inc}}$. The distribution of the tangential $\mathbf{E}$-field of the υ mode at the ζ-port is described by the modal vector $\mathbf{e}_\zeta^{(\upsilon)}$, which can be obtained by means of 2-D port analysis. Using Maxwell's equations for a traveling wave, the dual modal vector $\mathbf{h}_\zeta^{(\upsilon)}$ is obtained. Note that the modal vectors are by definition tangential to the port. In the case of isotropic loss-free ports, they are real valued [20].

The pairs of the modal vectors $(\mathbf{e}_\zeta^{(\upsilon)}, \mathbf{h}_\zeta^{(\upsilon)})$ form a bi-orthogonal set and have to be suitably normalized. The common bi-orthonormal relation is:[4]

$$\iint_{S_\varsigma} \left(\mathbf{e}_\zeta^{(\upsilon)} \times \mathbf{h}_\varsigma^{(\upsilon')}\right) \cdot d\mathbf{s} = \begin{cases} 1, & \text{if } \upsilon = \upsilon' \\ 0, & \text{if } \upsilon \neq \upsilon'. \end{cases} \tag{34}$$

The jth-port cross section S_j is part of S_Ω and it defines the excitation port where the boundary condition (20) and (21) holds. The position of S_j along the jth waveguide can be adjusted virtually by port deembedding [3]. In general, $\mathbf{E}_{j\tan}^{\text{inc}}$ may be a superposition of modes

$$\mathbf{E}_{j\tan}^{\text{inc}} = \sum_m V_j^{(m)} \mathbf{e}_j^{(m)} \tag{35}$$

where $V_j^{(m)}$ indicates the magnitude of the mth mode incident on the jth-port. From (34), it follows that the denominator in (33) is the magnitude of the incident υ-mode wave

$$V_j^{(\upsilon)} = \iint_{S_j} (\mathbf{E}_j^{\text{inc}} \times \mathbf{h}_j^{(\upsilon)}) \cdot d\mathbf{s}. \tag{36}$$

In this analysis, we assume that the ports are independent of the optimizable parameters. $V_j^{(\upsilon)}$ and the modal vectors $\mathbf{h}_\zeta^{(\upsilon)}$, $\zeta = j, k$ are then independent of $p_n, n = 1, \ldots, N$. The derivative of S_{kj} is obtained as

$$\frac{\partial S_{kj}}{\partial p_n} = \frac{1}{V_j} \cdot \frac{\partial F_{kj}}{\partial p_n} \tag{37}$$

where

$$F_{kj} = \iint_{S_k} (\mathbf{E}_j \times \mathbf{h}_k) \cdot d\mathbf{s}. \tag{38}$$

Hereafter, the superscript (υ) is omitted for simpler notations.

The comparison of F_{kj} with the general response definition in (1) reveals that, in this case,

$$f = 0 \tag{39}$$

$$f_s = \begin{cases} (\mathbf{E}_j \times \mathbf{h}_k) \cdot \mathbf{a}_n, & \text{at } S_k \\ 0, & \text{elsewhere on } S_\Omega. \end{cases} \tag{40}$$

From (19) and (39), it follows that the impressed volumetric adjoint excitation is zero, $\hat{\mathbf{G}}^i = 0$. The adjoint field ($\hat{\mathbf{E}}_n$ or $\hat{\mathbf{E}}$) satisfies the respective source-free equations (18) or (28). We may consider only $\hat{\mathbf{E}}$ since the adjoint fields $\hat{\mathbf{E}}_n$ $(n = 1, \ldots, N)$ for shape sensitivities are derived from it. Note that the field solutions $\bar{\mathbf{E}}_j$ $(j = 1, \ldots, K)$ also satisfy the source-free equation of the nominal state (10).

At the same time, as follows from (40), $f_s = (\mathbf{h}_k \times \mathbf{a}_n) \cdot \mathbf{E}_j$ at the kth-port. Compared with (26), we find the adjoint excitation vector at the kth-port as

$$\hat{\mathbf{U}}_k^{\text{inc}} = \frac{\mu_r (\mathbf{h}_k \times \mathbf{a}_n)}{\Delta h^2}. \tag{41}$$

At all other ports, the adjoint excitation is zero.

To express $\hat{\mathbf{U}}_k^{\text{inc}}$ in terms of the electric modal vector $\mathbf{e}_k$, the general termination condition at the kth-port is applied as follows:

$$\mathbf{a}_{n,k} \times (\nabla \times \mathbf{e}_k) + P(\mathbf{e}_k) = 0. \tag{42}$$

Since $\nabla \times \mathbf{e}_k = -j\omega\mu \mathbf{h}_k$, the dual modal vector h_k is found as

$$\mathbf{h}_k \times \mathbf{a}_{n,k} = -\frac{P(\mathbf{e}_k)}{j\omega\mu}. \tag{43}$$

[4]This orthogonality relation is the basis of the S-parameter definition in (33). It is valid for waveguides whose cross section contains heterogeneous isotropic medium; see, e.g., [22], [23]. The mode orthogonality in heterogeneous anisotropic guides is discussed in [23] and [24].

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Therefore,

$$\hat{\mathbf{U}}_k^{\text{inc}} = -\frac{P(\mathbf{e}_k)}{j\omega\mu_0\Delta h^2}. \tag{44}$$

At the same time, in the original simulation, in order to obtain all S_{ik} $(i = 1, \ldots, K)$ parameters, the modal excitation $\mathbf{U}_k^{\text{inc}} = 2P(\mathbf{E}_k^{\text{inc}})$ [see (21)] is applied, where $\mathbf{E}_k^{\text{inc}} = V_k\mathbf{e}_k$. This is the simulation generating the field solution $\bar{\mathbf{E}}_k$. In view of the linearity of the operator P, it follows that

$$\hat{\mathbf{U}}_k^{\text{inc}} = 2V_k P(\mathbf{e}_k). \tag{45}$$

The comparison of (44) and (45) shows that $\hat{\mathbf{U}}_k^{\text{inc}}$ and $\mathbf{U}_k^{\text{inc}}$ relate through a complex-scalar constant, and thus, have common modal distribution. Since $\bar{\mathbf{E}}_k$ and $\hat{\mathbf{E}}$ satisfy the same system equation, the linearity of the problem suggests that their ratio at every point of the computational domain is equal to the ratio of their excitations, i.e.,

$$\hat{\mathbf{E}} = \kappa_{kj}\bar{\mathbf{E}}_k \quad \kappa_{kj} = -(2V_k j\omega\mu_0\Delta h^2)^{-1}. \tag{46}$$

The constant κ_{kj} is referred to as the self-adjoint constant. It allows to obtain the adjoint field $\hat{\mathbf{E}}$ from $\bar{\mathbf{E}}_k$ without performing the adjoint system analysis. Thus, the sensitivity formula (27) for the S-parameters can now be written without any reference to an adjoint field

$$\frac{\partial S_{kj}}{\partial p_n} \approx \frac{1}{2V_k V_j j\omega\mu_0\Delta h^2} \iiint_{\Omega} (\bar{\mathbf{E}}_k)_n \cdot \frac{\Delta_n R(\bar{\mathbf{E}}_j)}{\Delta p_n} d\Omega. \tag{47}$$

Notice that the explicit derivative $\partial^e S_{kj}/\partial p_n$ has been set to zero. This is due to the aforementioned assumption that the ports do not depend on the optimizable parameters. The self-adjoint analytic sensitivity formula derived from (29) is analogous, i.e., the constant factor multiplying its integral term is the same as in (47).

In summary, the S-parameter sensitivity computation based on (47) can be carried out as a post-process. It requires: 1) the complex vector field values $\bar{\mathbf{E}}_i$ $(i = 1, \ldots, K)$ at the perturbation grid points and 2) the modal amplitudes V_i $(i = 1, \ldots, K)$ at all ports. The latter are not always available if commercial simulation packages are used. However, in many cases, they can be obtained indirectly. For example, for S-parameter analysis, the FEM simulators carry out a preliminary 2-D port analysis [25] at all ports in order to set $\mathbf{E}_i^{\text{inc}}$ $(i = 1, \ldots, K)$ as a single desired mode. The procedure is necessary to set up the port boundary conditions. From these analyses, the port impedances Z_i are often available. In the case of a loss-free port, V_i can then be obtained as $V_i = \sqrt{Z_i P_i^{\text{inc}}}$ where P_i^{inc} is the incident power at the port. The latter is usually user defined or set to unity by default.

V. S-Parameter Sensitivities in Design Examples

Our self-adjoint S-parameter sensitivity analysis technique based on the FDFD system equations has been already verified extensively [11], [14], [16], [17] by comparisons with response-level finite-difference approximations of the S-parameter derivatives. These are not repeated here. They show that the accuracy of the self-adjoint derivatives is comparable or better than that of the central finite-difference approximations. However, they are more reliable since there is no ambiguity associated with the choice of the parameter perturbation Δp_n. The self-adjoint derivative computation is orders of magnitude faster than the response-level estimates because it does not involve full-wave simulations.

The proposed approach applies its own finite-difference grid, which is fitted locally to each of the optimizable objects [14]. This makes it independent of the simulator's grid and system matrices. The independent sensitivity-analysis grid in general requires interpolation of the field solution in order to transfer it from the simulation grid (e.g., an FEM tetrahedral mesh). In our examples, we use a commercial FEM solver [25], which provides such interpolation utility for field export through the "field calculator."

Here, we focus on the application of the S-parameter sensitivities to gradient-based optimization, which is a basic tool in design tuning. We demonstrate the significant time savings achieved when our method provides the Jacobian to the optimization algorithm instead of using response-level finite-difference estimates. We demonstrate optimization of structures where both dielectric and metallic objects are considered, including infinitesimally thin metallic objects.

A. Minimax Optimization

The design problem is defined as

$$p^* = \arg\min_{\boldsymbol{p}} F(\boldsymbol{R}(\boldsymbol{p})) \tag{48}$$

where $\boldsymbol{R}$ is the vector of M responses, $\boldsymbol{p}$ is the vector of design parameters, and $\boldsymbol{p}^*$ is the optimal solution, which minimizes the objective function F. In our examples, the M responses are the S-parameter magnitudes evaluated at select frequencies where design goals specify lower (L) or upper (U) bounds for the responses. We use a minimax objective function defined as $F = \max\{e_1, \ldots, e_M\}$. The errors e_m $(m = 1, \ldots, M,\ M = M_L + M_U)$ are: 1) for a lower bound, $e_m = L_m - R_m$ $(m = 1, \ldots, M_L)$ and 2) for an upper bound, $e_m = R_m - U_m$, $(m = M_L + 1, \ldots, M)$.

B. Optimization of an H-Plane Filter

The six-resonator H-plane filter [26] is shown in Fig. 4. The rectangular waveguide is of width $2a = 34.85$ mm and height $b = 15.80$ mm. The six resonators are separated by seven septa of finite thickness $\delta = 0.6223$ mm. The design parameters are the septa lengths L_1, L_2, L_3, and L_4. The resonator lengths are fixed at $S_1 = S_2 = 16.18$ mm and $S_3 = 16.80$ mm.

We use Madsen's minimax optimization algorithm [27], which employs a trust region and requires the Jacobian $\nabla_p F$. The minimax objective function uses the specifications

$$\begin{aligned} &|S_{11}| \geq 0.85 \ f \leq 5.2 \text{ GHz} \\ &|S_{11}| \leq 0.16 \ 5.4 \text{ GHz} \leq f \leq 9 \text{ GHz} \\ &|S_{11}| \geq 0.5 \ f \geq 9.5 \text{ GHz}. \end{aligned} \tag{49}$$

The S-parameters and their Jacobians are calculated at 22 frequency points from 5.0 to 10.0 GHz. The initial design is given by $\boldsymbol{p}^{(0)} = [L_1\ L_2\ L_3\ L_4]^T = [5\delta\ 11\delta\ 9\delta\ 13\delta]^T$. The

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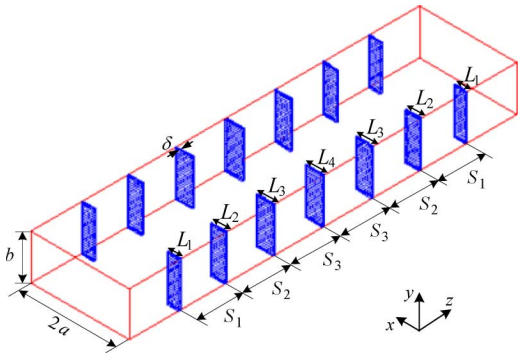


Fig. 4. Six-resonator H-plane filter [26].

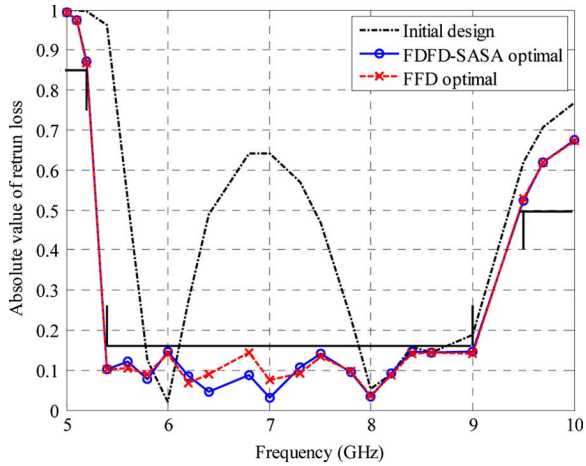


Fig. 5. $|S_{11}|$ of the H-plane filter versus frequency in the initial and the optimal designs. Upper and lower specification bounds are indicated by solid lines.

initial trust-region radius is set to $r^{(0)} = 0.05 \cdot \|\boldsymbol{p}^{(0)}\|$.[5] The optimization process is terminated when one of the following conditions is fulfilled for two consecutive iterations: $\Delta_p^{(k)} = \|\boldsymbol{p}^{(k+1)} - \boldsymbol{p}^{(k)}\| / \|\boldsymbol{p}^{(k)}\| \leq 0.001$, $\Delta_F^{(k)} = \|F(k+1) - F(k)\| \varepsilon$, where $\varepsilon = 0.009$ is the convergence error of the mesh refinement in the simulation.

The purpose here is to examine the impact of our proposed sensitivity-analysis approach on the efficiency of the overall optimization process. This is why we perform two separate optimization processes, which use the same optimization algorithm, the same initial point, the same initial trust region radius, and the same stopping criterion. They differ only by the method used to compute the gradient $\nabla_{\boldsymbol{p}} F$.

In the first optimization procedure, we use the proposed method to calculate the S-parameter Jacobians from which $\nabla_{\boldsymbol{p}} F$ is obtained. The assumed central-node FDFD grid is uniform with cell size $\Delta h = \delta = 0.6223$ mm. All respective curves in the following plots are denoted by "FDFD-SASA."

In the second optimization procedure, we use response-level forward finite differences to calculate $\nabla_{\boldsymbol{p}} F$. The results thus obtained are marked by "FFD." With this method, one has to carefully choose the parameter perturbations Δp_n, $n = 1, \ldots, N$. Large perturbations may miss local nonlinear behavior. Too small perturbations may result in a difference between the objective functions of the nominal and perturbed structures, which is well below the accuracy of the numerical solution. In such a case, the finite-difference estimate of the Jacobian is erroneous. After some trials, we found that a perturbation of 1% for all design parameters is acceptable, i.e., the agreement between the forward and backward finite-difference estimates is good at the initial design.

Fig. 5 shows the reflection $|S_{11}|$ of the initial design and the optimal designs obtained using the two different optimization processes. Fig. 6 shows the objective functions versus the optimization iterations. It is observed that both optimization processes converge within ten iterations with comparable objective-function values: the "FDFD-SASA" optimization converges with $F = -0.01569$, while the "FFD" optimization converges with $F = -0.01381$. Yet it is clear that the two optimization processes are different in both the evolution of the objective function F and the final design. This is because the Jacobian obtained using the proposed method and that estimated with the forward finite differences are different.

The impact of the method for calculating the Jacobian on the overall CPU time of the optimization process is shown in Table II. Despite the equal number of iterations, the "FFD" optimization process takes much longer time than the "FDFD-SASA" one. This is because with forward finite differences, the Jacobian calculation needs one additional simulation per optimizable parameter (the perturbed-structure simulation) where the particular parameter is perturbed by 1% of its nominal value while all other parameters are kept at their nominal values. In this example, there are $N = 4$ optimizable parameters. Thus, at each optimization iteration, five simulations are needed—one for the nominal state and four more for the perturbed states. At the same time, the "FDFD-SASA" approach needs only one simulation per iteration—that of the nominal state. The overhead of computing the sensitivity integrals (47) is less than 1 s, and is thus, negligible. The major overhead that we encounter is actually associated with exporting the $\mathbf{E}$-field solution at the perturbation grid points. For all ten iterations, this amounts to about 1536 s. Even with this overhead, the "FDFD-SASA" minimax optimization is roughly 14 times faster than the "FFD" alternative.

C. Optimization of a Dielectric-Resonator Antenna

In a second example, we consider a dielectric-resonator antenna [28] fed by a microstrip line through an aperture in the ground plane, which is assumed infinitesimally thin (see Fig. 7). The rectangular dielectric resonator has its bottom face centered onto the aperture.

The optimizable parameters are: the resonator permittivity ε_{r1}, the three shape parameters of the resonator a, b, and c, and the shape parameter of the feed aperture l_a. The initial design is set as $\boldsymbol{p}^{(0)} = [\varepsilon_{r1}\ a\ b\ c\ l_a]^T = [12\ 23\ 23\ 10\ 20]^T$. The shape parameters are in millimeters. The design specifications are

$$\begin{aligned} |S_{11}| &\geq 0.65 \quad 2.0 \leq f \leq 2.4 \text{ GHz} \\ |S_{11}| &\leq 0.2 \quad 2.82 \leq f \leq 2.98 \text{ GHz} \\ |S_{11}| &\geq 0.65 \quad 3.4 \leq f \leq 3.8 \text{ GHz}. \end{aligned} \tag{50}$$

[5]The initial trust-region radius depends on the nonlinearity of the objective function at the initial point. Most algorithms use an *ad hoc* value provided by the user. This is the case with the minimax algorithm here. The value above was chosen so that it is sufficiently smaller than $0.1 \cdot \|\boldsymbol{p}^{(0)}\|$—a value recommended for a weakly nonlinear problem.

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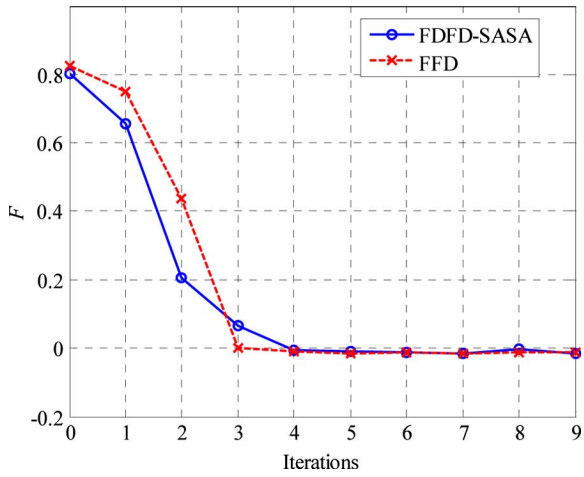


Fig. 6. Objective function versus the iterations in the optimization of the H-plane filter. The optimal parameter values are (all in millimeters): $\boldsymbol{p}^*_{\text{FDFD-SASA}} = [4.4680\ 5.3042\ 5.9099\ 5.9024]^T$ and $\boldsymbol{p}^*_{\text{FFD}} = [4.3732\ 5.3892\ 5.8185\ 5.9821]^T$.

TABLE II
NUMBER OF ITERATIONS AND TIME COMPARISON BETWEEN DIFFERENT SENSITIVITY ANALYSIS METHODS*

	FFD	FDFD-SASA
Optimization iterations	10	10
Calls for EM simulation	50	10
Response estimation (s)	5370	5370
Jacobian estimation (s)	21480	0.31

*Ansoft HFSS is run on Intel@ Pentium@4 with CPU 3 GHz, RAM 1 GB

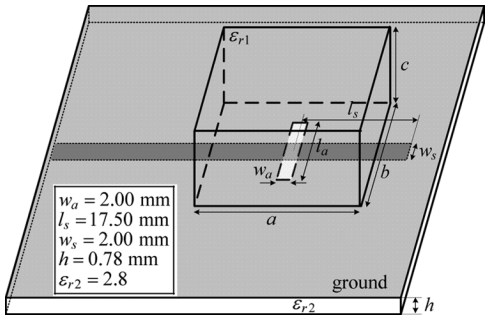


Fig. 7. Dielectric-resonator antenna [28].

The same minimax algorithm is used as in the previous example. The initial trust region radius is set as $r^{(0)} = 0.01 \cdot \|\boldsymbol{p}^{(0)}\|$. The termination criterion employs the conditions $\Delta_p^{(k)} \leq \delta$ or $\Delta_F^{(k)} \leq \varepsilon$, for two consecutive iterations, where $\delta = 0.013$ and $\varepsilon = 0.01$.

As in the previous example, the parameter perturbations used for the "FFD" Jacobian estimates are selected carefully. The following values are used: $\Delta\boldsymbol{p}_{\text{FFD}} = [0.5\ \ 0.04a\ \ 0.04b\ \ 0.05c\ \ 0.05l_a]^T$. The adjoint sensitivity calculation uses different grid step sizes Δx, Δy, and Δz for the different parameters. They are always chosen so that they are roughly equal to the FEM mesh edge lengths along the respective dimension of the optimized object.

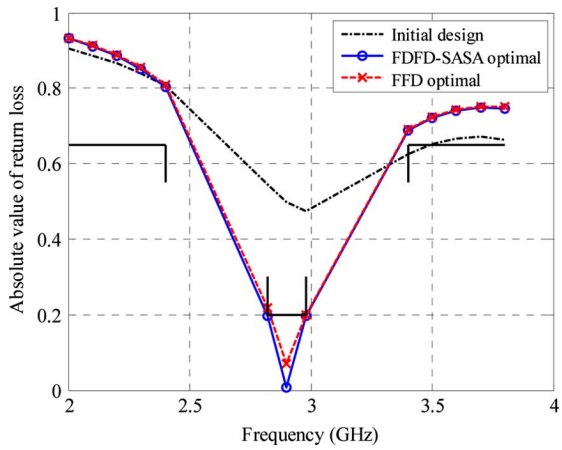


Fig. 8. $|S_{11}|$ of the dielectric-resonator antenna versus frequency in the initial and optimal designs.

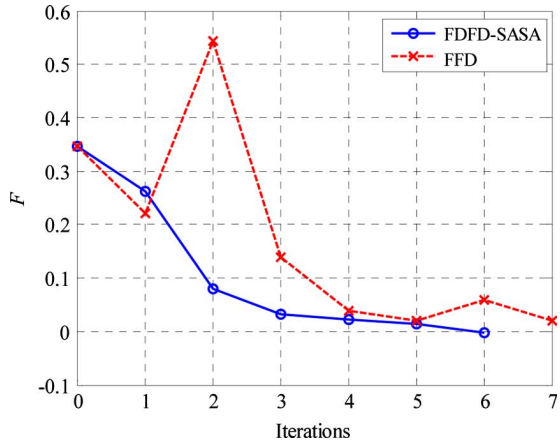


Fig. 9. Objective functions versus the iterations in the optimization of the antenna. The optimal parameter values are $\boldsymbol{p}^*_{\text{FDFD-SASA}} = [10.06\ \ 23.66\ \ 23.47\ \ 13.64\ \ 17.46]^T$ and $\boldsymbol{p}^*_{\text{FFD}} = [10.59\ 23.58\ 23.21\ 12.74\ 17.21]^T$.

Fig. 8 shows the reflection coefficient of the initial design, as well as the final optimal designs obtained in the two separate optimization processes. The two final designs show comparable performance; however, the "FFD" final design violates the design specifications at 2.98 GHz. Fig. 9 shows the evolution of the objective functions of the two processes versus iterations. This plot shows that the two optimization processes evolve differently. This suggests that the Jacobians supplied by the two sensitivity-analysis methods differ significantly. The "FDFD-SASA" optimization process stops because of the second condition in the termination criterion, i.e., $\Delta_F^{(k)} \leq 0.01$. The corresponding optimal design meets the specifications. On the other hand, the "FFD" optimization process ends due to the first termination condition $\Delta_p^{(k)} \leq 0.013$ with a positive objective function $F = 0.0195$, i.e., the design specifications are not met.

The value of δ in the first termination condition is usually chosen so that it is less than the perturbation used by the "FFD" Jacobian estimation (roughly 4% in this example) and it corresponds to a change of a shape parameter, which is roughly equal to the shortest mesh edge length found at the optimizable object. Choosing δ much smaller usually does not improve the

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convergence of the optimization process with "FFD" gradients because taking very small steps in the optimizable parameter space leads to practically identical responses whose difference is below ε.

To confirm the above statement, we repeat the two optimization processes with only one change in the termination criterion, namely, we set δ an order of magnitude smaller as $\delta = 0.0013$. The "FDFD-SASA" design optimization proceeds in a manner identical to the one shown in Fig. 9. The "FFD" process again fails to converge. After 12 iterations, it terminates with $F = 0.0077$. Again, the condition $\Delta_p^{(k)} \leq \delta$ terminates the optimization process. This example illustrates the fact that the poor accuracy of the "FFD" Jacobian estimates may prevent the optimization process from converging to a solution, which actually exists and which can be found with the more accurate adjoint-based derivatives.

The time efficiency of the adjoint-based sensitivity calculation is evident in this antenna example as well. One FEM simulation (one frequency sweep) of this structure takes about 220 s.[6] A system analysis with the "FFD" Jacobians needs six FEM simulations, which requires approximately $220 \times 6 = 1320$ s of which $220 \times 5 = 1100$ s are due to the Jacobian computation. At the same time, a system analysis, where the Jacobians are computed with the proposed method, only one FEM simulation is carried out from which the S-parameters and their Jacobians are obtained. The time required to compute the Jacobians is about 0.1 s, which is more than four orders of magnitude reduction compared to the "FFD" estimation.

In summary, the proposed approach to Jacobian calculation offers two major advantages, which are: 1) it is orders of magnitude faster than the response-level finite differences and 2) it is more accurate, and thus, improves the convergence of simulation-based gradient optimization processes.

VI. Conclusion

We have proposed a method for the sensitivity analysis of S-parameters, which can be implemented as an independent software module producing the S-matrix derivatives from the complex-phasor $\mathbf{E}$-field solutions. The method is based on the system equations of the FDFD technique applied to the $\mathbf{E}$-field Helmholtz equation and discretized on a central-node finite difference grid. To keep the derivation concise, some simplifying assumptions have been made, which are: 1) the medium is isotropic and 2) the ports do not depend on the optimizable parameters. Anisotropic cases, however, can be solved with the self-adjoint formulation provided the material tensors are symmetric or Hermitian. The ports can also be included by taking into account the explicit derivatives related to the ports' shape, materials, and modal vectors.

In realistic design examples, we demonstrate the time efficiency of our Jacobian computations whose overhead is negligible compared with the time required by the full-wave simulation. We demonstrate how this efficiency leads to significant reduction of the time required by design tuning where gradient-based optimization algorithms are employed. Finally, we show that the convergence of gradient-based optimization procedures improves due to the superior accuracy of our method as compared to response-level finite differences or preliminary parameter sweeps. The latter approaches appear to be not only extremely time inefficient, but also unreliable. The proposed sensitivity solver has the potential to dramatically improve the optimization capabilities of current high-frequency CAD packages and to lead to a whole range of new optimization approaches, which take advantage of the response Jacobians.

The accuracy of the proposed method has been extensively evaluated for objects whose shapes conform to the assumed finite-difference grid [11], [14], [16], [17], [29]. It has been shown that, in such cases, the accuracy is superior to the second-order response-level estimates. Yet the finite-difference grid is inherently inadequate in representing curved surfaces and slanted planar surfaces, which do not conform to the grid planes. Staircase approximations are certainly possible. However, more sophisticated approaches can be employed such as those developed for curvilinear coordinate systems and for the conformal-grid finite-difference time-domain method [30], [31]. The method described here is general and allows for integration with any of the known conformal techniques for finite-difference grids. The accuracy of such conformal finite-difference grid SASA is yet to be thoroughly examined and quantified.

Appendix

Consider a waveguide of cross section where the medium may in general be characterized by the permittivity and permeability tensors $\boldsymbol{\varepsilon}$ and $\boldsymbol{\mu}$, both of which may be functions of the cross-section coordinates. Consider two sets of fields, the field propagating in the original waveguide

$$\begin{aligned} \nabla \times \mathbf{E} &= -j\omega\boldsymbol{\mu} \cdot \mathbf{H} \\ \nabla \times \mathbf{H} &= j\omega\boldsymbol{\varepsilon} \cdot \mathbf{E}. \end{aligned} \tag{51}$$

and an adjoint field propagating in the "adjoint" guide where the material tensors are the transposed of the original ones

$$\begin{aligned} \nabla \times \hat{\mathbf{E}} &= -j\omega\boldsymbol{\mu}^T \cdot \hat{\mathbf{H}} \\ \nabla \times \hat{\mathbf{H}} &= j\omega\boldsymbol{\varepsilon}^T \cdot \hat{\mathbf{E}}. \end{aligned} \tag{52}$$

Assume that both fields propagate along $+z$ according to $e^{-\gamma z}$ and $e^{-\hat{\gamma} z}$, respectively. Following a procedure identical to that in [22], the following identity is obtained at a cross-section S_i of the guide

$$\begin{aligned} \oint_{C_i} \left[(\mathbf{a}_z \times \mathbf{E}) \cdot \hat{\mathbf{H}} - (\mathbf{a}_z \times \hat{\mathbf{E}}) \cdot \mathbf{H} \right] dl \\ = (\gamma + \hat{\gamma}) \iint_{S_i} \left(\mathbf{E} \times \hat{\mathbf{H}} - \hat{\mathbf{E}} \times \mathbf{H} \right) \cdot \mathbf{a}_z ds \end{aligned} \tag{53}$$

[6]CPU time recorded on an Intel Xeon 5160 dual-core computer with CPU 3 GHz and RAM 16 GB.

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Here, C_i is the contour bounding S_i. The contour integral over C_i vanishes regardless of whether it belongs to a PEC, PMC, or an open boundary at infinity. Since $\gamma \neq -\hat{\gamma}$,[7] the identity

$$\iint_{S_i} (\mathbf{E} \times \hat{\mathbf{H}}) \cdot \mathbf{a}_z ds = \iint_{S_i} (\hat{\mathbf{E}} \times \mathbf{H}) \cdot \mathbf{a}_z ds \tag{54}$$

holds.

Further, the general boundary condition for the two traveling waves [see (20)] with $\mathbf{a}_z = \mathbf{a}_n$ can be written as

$$\begin{aligned} \mathbf{a}_z \times (j\omega \boldsymbol{\mu} \cdot \mathbf{H}) &= P(\mathbf{E}) \\ \mathbf{a}_z \times (j\omega \boldsymbol{\mu}^T \cdot \hat{\mathbf{H}}) &= P(\hat{\mathbf{E}}). \end{aligned} \tag{55}$$

These relations allow for the magnetic field vectors $\mathbf{H}$ and $\hat{\mathbf{H}}$ to be expressed in terms of their electric counterparts and then substituted in (54). The result is

$$\iint_{S_i} \mathbf{E}_{\tan} \cdot \left[\boldsymbol{\mu}^{-1} \cdot P(\hat{\mathbf{E}})\right] ds = \iint_{S_i} \hat{\mathbf{E}}_{\tan} \cdot \left[(\boldsymbol{\mu}^T)^{-1} \cdot P(\mathbf{E})\right] ds. \tag{56}$$

Since the above identity applies to both the solution of the nominal state $\mathbf{E}$ and the solution of the nth perturbed state $\mathbf{E}_n$, it can also be stated that

$$\begin{aligned} \iint_{S_i} \frac{\partial \mathbf{E}_{\tan}}{\partial p_n} \cdot \left[\boldsymbol{\mu}^{-1} \cdot P(\hat{\mathbf{E}})\right] ds \\ = \iint_{S_i} \hat{\mathbf{E}}_{\tan} \cdot \left[(\boldsymbol{\mu}^T)^{-1} \cdot P\left(\frac{\partial \mathbf{E}}{\partial p_n}\right)\right] ds. \end{aligned} \tag{57}$$

In the case of an isotropic medium, (57) reduces to (24).

REFERENCES

[1] H. Akel and J. P. Webb, "Design sensitivities for scattering-matrix calculation with tetrahedral edge elements," *IEEE Trans. Magn.*, vol. 36, no. 4, pp. 1043–1046, Jul. 2000.

[2] M. H. Bakr, N. K. Nikolova, and P. A. W. Basl, "Self-adjoint S-parameter sensitivities for lossless homogeneous TLM problems," *Int. J. Numer. Modeling*, vol. 18, no. 6, pp. 441–455, Nov./Dec. 2005.

[3] N. K. Nikolova, J. Zhu, D. Li, M. H. Bakr, and J. W. Bandler, "Sensitivity analysis of network parameters with electromagnetic frequency-domain simulators," *IEEE Trans. Microw. Theory Tech.*, vol. 54, no. 2, pp. 670–681, Feb. 2006.

[4] N. K. Nikolova, Y. Li, Y. Li, and M. H. Bakr, "Sensitivity analysis of scattering parameters with electromagnetic time-domain simulators," *IEEE Trans. Microw. Theory Tech.*, vol. 54, no. 4, pp. 1598–1610, Apr. 2006.

[5] P. A. W. Basl, M. H. Bakr, and N. K. Nikolova, "Theory of self-adjoint S-parameter sensitivities for lossless nonhomogeneous transmission-line modeling problems," *IET Microw. Antennas Propag.*, vol. 2, no. 3, pp. 211–220, Apr. 2008.

[6] L. Vardapetyan, J. Manges, and Z. Cendes, "Sensitivity analysis of S-parameters including port variations using the transfinite element method," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Atlanta, GA, Jun. 2008, pp. 527–530.

[7] F. Azadivar, "A tutorial on simulation optimization," in *Proc. ACM 24th Winter Simulation Conf.*, Arlington, VA, Dec. 1992, pp. 198–204.

[8] E. J. Haug, K. K. Choi, and V. Komkov, *Design Sensitivity Analysis of Structural Systems*. Orlando, FL: Academic, 1986.

[9] N. K. Nikolova, J. W. Bandler, and M. H. Bakr, "Adjoint techniques for sensitivity analysis in high-frequency structure CAD," *IEEE Trans. Microw. Theory Tech.*, vol. 52, no. 1, pp. 403–419, Jan. 2004.

[10] Q. Fang, P. M. Meaney, S. D. Geimer, A. V. Streltsov, and K. D. Paulsen, "Microwave image reconstruction from 3-D fields coupled to 2-D parameter estimation," *IEEE Trans. Med. Imag.*, vol. 23, no. 4, pp. 475–484, Apr. 2004.

[11] X. Zhu, A. Hasib, and N. K. Nikolova, "Electromagnetic sensitivity analysis of scattering parameters based on the FDFD method," in *Int. Signals, Syst., Electron. Symp.*, Montreal, QC, Canada, Jul.-Aug. 2007, pp. 165–168.

[12] K. Yee, "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media," *IEEE Trans. Antennas Propag.*, vol. AP-14, no. 5, pp. 302–307, May 1966.

[13] G. Veronis, R. W. Dutton, and S. Fan, "Method for sensitivity analysis of photonic crystal devices," *Opt. Lett.*, vol. 29, no. 19, pp. 2288–2290, Oct. 2004.

[14] X. Zhu, A. Hasib, N. K. Nikolova, and M. H. Bakr, "Efficient electromagnetic optimization using self-adjoint Jacobian computation based on a central-node FDFD method," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Atlanta, GA, Jun. 2008, pp. 979–982.

[15] N. K. Nikolova, H. W. Tam, and M. H. Bakr, "Sensitivity analysis with the FDTD method on structured grids," *IEEE Trans. Microw. Theory Tech.*, vol. 52, no. 4, pp. 1207–1216, Apr. 2004.

[16] A. Hasib, "Sensitivity analysis for design optimization of metallic microwave structures with the finite-difference frequency-domain method," M.A.Sc., Dept. Elect. Comput. Eng., McMaster Univ, Hamilton, ON, Canada, 2008.

[17] X. Zhu, "Frequency-domain self-adjoint S-parameter sensitivity analysis for microwave design," M.A.Sc thesis, Dept. Elect. Comput. Eng., McMaster Univ., Hamilton, ON, Canada, 2008.

[18] W. C. Chew, M. S. Tong, and B. Hu, *Integral Equation Methods for Electromagnetic and Elastic Waves*. San Rafael, CA: Morgan & Claypool, 2008, pp. 24–39.

[19] J. Jin, *The Finite Element Method in Electromagnetics*, 2nd ed. New York: Wiley, 2002, pp. 309, 348–355.

[20] M. Salazar-Palma, T. K. Sarkar, L.-E. García-Castillo, T. Roy, and A. Djordjevic, *Iterative and Self-Adaptive Finite-Elements in Electro-magnetic Modeling*. Norwood, MA: Artech House, 1998, pp. 462–463, 465–466.

[21] M. H. Bakr and N. K. Nikolova, "An adjoint variable method for frequency domain TLM problems with conducting boundaries," *IEEE Microw. Wireless Compon. Lett.*, vol. 13, no. 9, pp. 408–410, Sep. 2003.

[22] R. E. Collin, *Field Theory of Guided Waves*. New York: IEEE Press, 1991, p. 417.

[23] A. T. Villeneuve, "Orthogonality relationships for waveguides and cavities with inhomogeneous anisotropic media," *IRE Trans. Microw. Theory Tech.*, vol. MTT-7, no. 4, pp. 441–446, Oct. 1959.

[24] P. R. McIsaac, "Mode orthogonality in reciprocal and nonreciprocal waveguides," *IEEE Trans. Microw. Theory Tech.*, vol. 39, pp. 1808–1816, Nov. 1991.

[25] Ansoft High Frequency Structure Simulator (HFSS). ver. 11, Ansoft Corporation, Pittsburgh, PA, 2006. [Online]. Available: www.ansoft.com

[26] G. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*. Norwood, MA: Artech House, 1980, pp. 545–547.

[27] J. W. Bandler, W. Kellerman, and K. Madsen, "A superlinearly convergent minimax algorithm for microwave circuit design," *IEEE Trans. Microw. Theory Tech.*, vol. MTT-33, no. 12, pp. 1519–1530, Dec. 1985.

[28] D. Yau and N. V. Shuley, "A moment method analysis of an aperture coupled rectangular dielectric resonator antenna," in *IEEE AP-S/URSI Int. Antennas Propag. Symp.*, Jul. 1999, vol. 2, pp. 858–861.

[29] X. Zhu and N. K. Nikolova, "Accuracy improvement of the S-parameter adjoint sensitivity analysis for shape parameters," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Boston, MA, Jun. 2009, to be published.

[30] A. Taflove and S. C. Hagness, *Computational Electrodynamics: The Finite-Difference Time-Domain Method*, 2nd ed. Norwood, MA: Artech House, 2000, ch. 10.

[31] W. Yu and R. Mittra, *Conformal Finite Difference Time Domain Maxwell's Equations Solver, Software and User's Guide*. Boston, MA: Artech House, 2003.

[7] In fact, as is shown later, in the case of S-parameter analysis, $\gamma = \hat{\gamma}$.

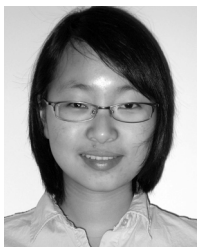
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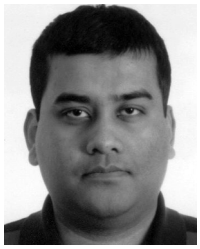
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